

Fluid Dynamics of the Atmosphere and Ocean, ECMM719, 2017.

Problem Set 1A (Assessed)

This problem set counts 10% toward your final mark. It is due on 24 February, 2017. Full marks can be obtained by doing 6 questions. All questions have equal credit, and the subsections within questions are also scored evenly.

Partial credit will be given if the algebra is wrong but the method right, *provided you explain what you are trying to do*. Mathematical communication counts for 10% of the marks.

1. (a) Consider a tracer in a fluid that obeys

$$\rho \frac{D\phi}{Dt} = -\nabla \cdot \mathbf{F},$$

where ρ is the density and $\mathbf{F}$ is the flux of ϕ (for example, $\mathbf{F} = -\kappa \nabla \phi$ so that $\nabla \cdot \mathbf{F} = -\kappa \nabla^2 \phi$ if κ is constant). If the fluid is contained in a box with solid walls and no flux of ϕ through the walls, show that

$$\int \rho \phi \, dV = \text{constant}.$$

- (b) Why is there no diffusion term in the mass conservation equation? That is, why is the mass continuity equation given by

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad \text{and not} \quad \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = \kappa \nabla^2 \rho, \quad (\text{MC.1})$$

with $\kappa \nabla^2 \rho$ representing the random motion of the molecules. Explain your answer with a brief derivation of the mass continuity equation, or otherwise.

2. In both parts to this question, make reasonable, approximate, assumptions about the size of the ocean, and state what they are. You do not need to be exact.

- (a) Using the observed value of molecular diffusion of heat in water, about $1.3 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, estimate how long it would take for a temperature anomaly to mix from the top of the ocean to the bottom, assuming that molecular diffusion alone is responsible. Comment on whether you think the real ocean has reached equilibrium after the last ice age (which ended about 10 000 years ago).
- (b) Suppose that water sinks near the pole over an area of $100 \times 100 \text{ km}$ (and so an area of 10^4 km^2), with a vertical velocity of 0.2 cm s^{-1} , and then spreads over the rest of the ocean. Estimate how long it would take this water to fill up the ocean basins (or, if you prefer, just the Atlantic). Comment on what this answer and your answer to part (a) tell us about how the deep ocean maintains its temperature.

3. (a) Consider a fluid that obeys the hydrostatic relation

$$\frac{\partial p}{\partial z} = -\rho g.$$

Suppose also that the fluid is an isothermal ideal gas. Show that the density and pressure both diminish exponentially with height. What is the e-folding height? (This is also called the ‘scale height’ of the atmosphere.) Write down an expression for the height, z , as a function of pressure.

- (b) Now suppose that the atmosphere has a uniform lapse rate (i.e., $dT/dz = -\Gamma = \text{constant}$). Show that the height at a pressure p is given by

$$z = \frac{T_0}{\Gamma} \left[1 - \left(\frac{p_0}{p} \right)^{-R\Gamma/g} \right]$$

where T_0 is the temperature at $z = 0$.

- (c) Are the answers you obtained in these two parts consistent in the isothermal (constant temperature) limit? Explain.

4. (a) Show that the vertical velocity within a shallow water system is given by

$$w = \frac{z - \eta_b}{h} \frac{Dh}{Dt} + \frac{D\eta_b}{Dt}.$$

Interpret this result, showing that it gives sensible answers at the top and bottom of the fluid layer. You may do the problem with $\eta_b = 0$.

- (b) In the shallow water equations show that, if the flow is approximately geostrophically balanced, the energy at large scales is predominantly potential energy and that energy at small scales is predominantly kinetic energy. Define precisely what ‘large scale’ and ‘small scale’ mean in this context, and obtain an expression for the transition scale.

5. The formula

$$\left(\frac{d\mathbf{A}}{dt} \right)_I = \left(\frac{d\mathbf{A}}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{A}$$

relates the total rate of change of a vector in an inertial frame (subscript I) to that in a frame rotating with angular velocity $\boldsymbol{\Omega}$ (subscript R). Obtain an expression for

$$\left(\frac{d^2\mathbf{A}}{dt^2} \right)_I$$

in terms of rotating frame quantities. If $\mathbf{A} = \mathbf{r}$, where $\mathbf{r}$ is the position vector, interpret your expression and identify the Coriolis and centrifugal terms.

Hence show that

$$\begin{aligned} \left(\frac{d^3\mathbf{A}}{dt^3} \right)_I &= \left(\frac{d^3\mathbf{A}}{dt^3} \right)_R + 3\boldsymbol{\Omega} \times \left(\frac{d^2\mathbf{A}}{dt^2} \right)_R \\ &\quad + 3\boldsymbol{\Omega} \times \left(\boldsymbol{\Omega} \times \left(\frac{d\mathbf{A}}{dt} \right)_R \right) - |\boldsymbol{\Omega}|^2 \mathbf{A}. \end{aligned}$$

6. (a) In an adiabatic shallow water fluid in a rotating reference frame show that the potential vorticity conservation law is

$$\frac{D}{Dt} \frac{\zeta + f}{\eta - h_b} = 0$$

where η is the height of the free surface and h_b is the height of the bottom topography, both referenced to the same flat surface.

- (b) An air column at 60° N with zero relative vorticity ($\zeta = 0$) stretches from the surface to the tropopause, which we assume is a rigid lid, at 10 km. The air column moves zonally on to a plateau 2.5 km high. What is its relative vorticity? Suppose it then moves southwards to 30° N, staying on the plateau. What is its relative vorticity then? (Assume that the density is constant.)

7. The shallow water equations, linearized about a state of rest, may be written as

$$\begin{aligned} \frac{\partial u'}{\partial t} - f_0 v' &= -g \frac{\partial \eta'}{\partial x}, & \frac{\partial v'}{\partial t} + f_0 u' &= -g \frac{\partial \eta'}{\partial y}, \\ \frac{\partial \eta'}{\partial t} + H \left(\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} \right) &= 0. \end{aligned}$$

Suppose there is a solid boundary (e.g., a coastline) at $x = 0$, with the ocean on one side and land on the other. Look for solutions that have $u' = 0$ everywhere, and with $f_0 > 0$. Show that the resulting waves are non-dispersive and travel at a speed $c = \sqrt{gH}$. Is the coastline to the left or to the right of the direction of travel? Suppose that these waves are generated just off the shore of Los Angeles. Do they move north or south?